



Term Evaluation (Odd) Semester Regular/Back Examination September 2026

Roll no.2318292.....

Name of the Program : B.TECH CSE

Semester: IIIrd

Name of the Course: Logic Design & Computer Organization

Course Code: TCS 308

Time: 1.5 hours

Maximum Marks: 25

Note:

- (i) This QP has 3 sections. Attempt questions from each Section.
- (ii) Answer the questions as per the instructions in each section.

Section A

This section has MCQ's. Attempt any 5. Each question carries 1 mark.

1. A 6-variable K-map contains
a) 64 cells
b) 8 cells
c) 16 cells
d) 32 cells
CO-1
2. A full adder has
a) 1 input and 2 outputs
b) 2 inputs and 2 outputs
c) 3 inputs and 1 output
d) 3 inputs and 2 outputs
CO-1
3. A multiplexer having 8 input lines requires:
a) 2 select lines
b) 3 select lines
c) 4 select lines
d) 8 select lines
CO-1
4. A 4-to-16 decoder has
a) 4 inputs and 8 outputs
b) 8 inputs and 4 outputs
c) 4 inputs and 16 outputs
d) 16 inputs and 4 outputs
CO-1
5. A combinational logic circuit differs from a sequential logic circuit because its output depends upon:
a) Present input only
b) Previous state only
c) Present input and previous state
d) Clock frequency only
CO-2
6. Which flip-flop eliminates the invalid state of the SR flip-flop??
a) D flip-flop
b) JK flip-flop
c) T flip-flop
d) Master-Slave D flip-flop
CO-2



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7. Which type of register allows data to be shifted in and out serially? CO-2
a) Parallel-In Parallel-Out (PIPO)
b) Parallel-In Serial-Out (PISO)
c) Serial-In Serial-Out (SISO)
d) Both b) and c)
8. In a JK flip-flop, when $J = 0$ and $K = 1$, the output CO-2
a) Remains unchanged
b) Is reset to 0
c) Is set to 1
d) Toggles

Section B

Attempt any 2 question out of given 4 questions. Each question carries 5 marks.

Q1. Minimize the following Boolean function using a 5-variable K-map (CO1)
 $F(A,B,C,D,E) = \sum m(0,1,4,5,8,9,12,13,16,17,20,21,24,25,28,29)$ $F = \bar{D}$

Q2. Implement the full subtractor using Multiplexer. (CO1)

Q3. Design a Binary to BCD code converter circuit. (CO1)

Q4. Explain the 2-to-4 line decoder circuit with enable input. Construct a 4-to-16 line decoder with five 2-to-4 line decoders with enable. (CO1)

Section C

Attempt any 2 question out of given 4 questions. Each question carries 5 marks.

Q1. Explain the difference between latches and flip flops. Discuss the various triggering mechanism in a flip-flop. (CO2)

Q2. Convert D flip flop to T flip flop. (CO2)

Q3. Draw and explain the circuit of 4-bit Serial-In Parallel-Out (SIPO) shift register. (CO2)

Q4. Draw and explain the circuit of universal shift register. (CO2)